

FILM, Sweden

WMO: 024560

Lat: 60.2358N

Lon: 17.9043E

Elev: 35

StdP: 100.91

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-15.3	-20.8	0.6	-18.3	-17.2	0.8	-14.0	7.3	3.5	6.5	2.4	0.3	210	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	27.1	18.6	25.1	17.7	23.3	16.7	19.9	24.6	18.7	23.4	17.7	21.9	2.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.1	21.8	17.1	12.2	20.5	16.0	11.4	19.5	57.0	24.4	53.3	23.3	50.0	21.9	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.4	4.7	-22.1	29.4	4.0	1.7	-25.0	30.7	-27.3	31.7	-29.5	32.7	-32.4	33.9		
(5)			-22.5	20.8	4.0	1.5	-25.4	21.9	-27.7	22.8	-29.9	23.6	-32.8	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-3.5	-3.4	-0.6	4.5	9.4	13.6	16.8	15.7	11.2	5.6	1.5	-1.5		
	DBStd	8.19	5.33	5.19	4.65	3.78	3.76	3.25	3.02	2.92	3.28	3.70	3.85	5.04		
	HDD10.0	2132	418	376	327	168	58	6	0	1	26	142	254	356		
	HDD18.3	4591	677	609	586	414	277	145	67	91	214	393	504	614		
	CDD10.0	616	0	0	0	4	39	115	211	178	62	7	0	0		
	CDD18.3	33	0	0	0	0	1	3	20	9	0	0	0	0		
	CDH23.3	389	0	0	0	0	13	61	225	89	1	0	0	0		
	CDH26.7	65	0	0	0	0	1	7	47	11	0	0	0	0		
(14) Wind	WSAvg	2.1	2.1	2.1	2.2	2.3	2.4	2.1	2.0	1.9	2.0	2.1	2.1	2.1		
(15) Precipitation	PrecAvg	619	45	37	33	37	41	64	66	74	56	55	61	51		
	PrecMax	876	86	75	61	113	89	133	129	156	126	123	127	102		
	PrecMin	485	6	7	3	4	15	10	18	5	15	4	15	13		
	PrecStd	80	21	15	16	26	19	34	35	36	27	28	27	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	7.9	13.4	20.3	25.1	27.6	30.1	28.0	22.7	16.2	11.3	10.0		
		MCWB	4.9	5.0	7.5	11.7	15.6	17.7	19.2	19.4	16.3	12.8	10.1	8.3		
	2%	DB	5.3	5.9	10.3	17.2	22.2	25.1	27.9	25.8	20.5	14.0	9.6	7.6		
		MCWB	4.0	3.6	5.5	9.8	13.9	16.3	19.1	18.5	15.4	11.8	8.5	6.5		
	5%	DB	4.2	4.4	8.0	14.5	19.7	23.0	26.0	23.9	18.6	12.6	8.3	6.2		
		MCWB	3.2	2.8	4.5	8.5	12.6	15.3	18.6	17.4	14.6	10.8	7.3	5.2		
	10%	DB	3.0	3.0	5.9	11.9	17.3	20.9	23.8	22.1	16.9	11.3	7.1	4.8		
		MCWB	2.2	1.8	3.1	6.8	11.3	14.5	17.5	16.8	13.6	9.8	6.3	4.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	5.8	7.8	12.5	16.9	19.0	21.4	21.5	17.2	13.5	10.2	8.6		
		MCDB	6.3	7.4	12.4	19.2	22.9	25.3	26.3	25.6	21.1	15.3	10.9	9.7		
	2%	WB	4.2	4.1	6.1	10.3	14.9	17.5	20.3	19.6	16.0	12.1	8.7	6.7		
		MCDB	4.9	5.3	9.7	15.9	20.3	23.1	25.3	23.5	19.2	13.5	9.5	7.4		
	5%	WB	3.2	3.1	4.8	8.9	13.3	16.2	19.2	18.5	15.1	11.2	7.5	5.3		
		MCDB	4.0	4.3	7.3	13.9	18.7	21.4	24.2	22.3	18.0	12.4	8.2	6.0		
	10%	WB	2.2	1.9	3.5	7.4	11.8	15.1	18.2	17.4	14.1	10.0	6.5	4.0		
		MCDB	2.9	2.9	5.7	11.2	16.4	19.8	22.9	20.9	16.4	11.1	7.1	4.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	6.7	8.5	10.4	11.0	10.7	10.9	10.4	9.3	7.0	4.7	4.9		
		MCDBR	5.0	7.3	11.3	15.1	15.7	14.9	14.2	13.3	11.5	8.3	5.9	5.3		
	5% WB	MCWBR	4.5	5.6	7.2	8.6	8.8	7.5	6.6	6.7	7.0	6.3	5.3	4.8		
		MCDWR	4.6	6.3	10.4	13.7	13.7	12.6	12.1	11.1	10.3	7.3	5.7	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.271	0.313	0.333	0.367	0.366	0.357	0.382	0.374	0.341	0.328	0.295	0.380		
		taud	2.200	2.284	2.355	2.341	2.405	2.462	2.411	2.432	2.509	2.426	2.243	2.675		
	Ebn at Noon		497	681	795	830	860	875	839	817	788	664	458	355		
		Edh at Noon	40	67	85	103	104	100	103	94	74	58	39	28		
(44) All-Sky Solar Radiation	RadAvg		0.27	0.87	2.37	3.91	5.13	5.72	5.31	4.15	2.65	1.17	0.35	0.17		
	RadStd		0.04	0.15	0.27	0.37	0.47	0.39	0.52	0.40	0.30	0.19	0.08	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (46 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-110	N/A	N/A	N/A

Nomenclature: See separate page